

pPIRW: An Efficient and Accurate Precalculation Path Integral Random Walk Solver for Steady-State Thermal Simulation With Robin Boundary Conditions

Zhixuan Dong¹, Longlong Yang, Cuiyang Ding, Changhao Yan², *Member, IEEE*, Zhaori Bi³, *Member, IEEE*, Sheng-Guo Wang⁴, *Life Senior Member, IEEE*, Dian Zhou⁵, *Senior Member, IEEE*, and Xuan Zeng⁶, *Senior Member, IEEE*

Abstract—With the rapid increase of the transistor number in VLSI, rapidly rising power density and temperatures make heat dissipation a major challenge in IC design and manufacturing. However, conventional deterministic thermal analysis methods have difficulties in obtaining local temperature solutions efficiently, and the existing stochastic method is inaccurate when dealing with thermal analysis problems involving Robin boundary conditions (BCs). In this article, a highly parallelized path integral random walk (PIRW) solver is innovatively proposed for steady-state thermal analysis with mixed BCs, especially Robin BCs. The rigorous calculation of the local time and the Feynman–Kac functional $\hat{e}_c(t)$ are adopted to accurately handle Neumann and Robin BCs for the first time. Furthermore, based on the PIrw, we propose an accurate and microsecond-level precalculation PIrw (pPIRW) predictor, which precalculates time-consuming random walks, obtains temperatures by simple vector multiplication, and therefore is suitable for proactive thermal management. The pPIRW essentially calculates a partial inverse of large-scale matrices constructed from the finite difference-based compact thermal models (CTMs). Experimental results show that compared with 3D-ICE, the PIrw solver maintains high accuracy with a negligible error within 0.5 °C, achieves 136×–209× speedup and 8.53×–11.1× storage space reduction with all three kinds of BCs, and decreases to 1×–1.53× speedup for lacking the absorbing Dirichlet boundary. The pPIRW further has speed improvement of 2.5e4×–6.7e6× and memory reduction of 36.6×–42.5× over PIrw without loss of accuracy. Integrated within a thermal management strategy,

the pPIRW predictor can eliminate all thermal conflicts while maintaining the highest working frequency. Meanwhile, pPIRW achieves 29.2× speedup and 634× memory reduction over the CTM during the offline precalculation stage.

Index Terms—Feynman–Kac functional, random walk method, robin boundary conditions (BCs), temperature predictor, thermal analysis, thermal management.

I. INTRODUCTION

AS FEATURE sizes of process technologies scale down to the nanometer level, candidate technologies, such as 3-D ICs and chiplets are rapidly developed in order to meet the demand for continuous performance improvement. However, growing power density in advanced ICs causes serious thermal problems, threatening the performance, reliability, and aging of chips [1]. Thus, achieving precise and fast thermal simulation for ICs holds great significance, both during early stages of design and at runtime.

Thermal analysis is a significant subproblem of IC thermal issues, focusing on calculating temperature distributions by solving thermal conduction equations. There are mainly three kinds of thermal analysis methods: 1) deterministic; 2) neural network-based; and 3) stochastic methods. Deterministic methods involving full space discretization and solving linear equations are widely used for IC thermal analysis. Finite-element methods (FEMs), the ANSYS [2] and COMSOL [3] thermal simulators widely used in industry for example, discretize the whole domain, and then construct and solve the linear system. FEMs can attain very high accuracy through extremely fine discretization, and are flexible with complex structures [4]. Finite-difference methods (FDMs) [5], [6], such as HotSpot [7] and 3D-ICE [8], [9], [10] popular in academia, apply compact thermal models (CTMs) to discretize the domain into an equivalent thermal resistor–capacitor (RC) network of thermal nodes, and solve the RC equation. Nonetheless, they seem to be less flexible than FEMs for handling complex geometries and boundary conditions (BCs) [4]. These deterministic methods can obtain high accuracy at huge time and space costs, but they are applicable for neither the early stages of IC design where repeated thermal simulations are required, nor runtime temperature prediction where extremely fast thermal predictions are performed.

Received 8 February 2024; revised 16 August 2024; accepted 24 September 2024. Date of publication 3 October 2024; date of current version 21 March 2025. This work was supported in part by the National Key R&D Program of China under Grant 2020YFA0711900 and Grant 2020YFA0711904, and in part by the National Natural Science Foundation of China (NSFC) Research under Project 62474051, Project 62141407, Project 62474050, and Project 62304052. The preliminary version has been presented at ACM/IEEE International Conference on Computer-Aided Design (ICCAD) in 2022 [DOI: 10.1145/3508352.3549457]. This article was recommended by Associate Editor S. Vinco. (Corresponding authors: Changhao Yan; Xuan Zeng.)

Zhixuan Dong, Longlong Yang, Cuiyang Ding, Changhao Yan, Zhaori Bi, and Xuan Zeng are with the State Key Laboratory of Integrated Chips and Systems, Microelectronics Department, Fudan University, Shanghai 200433, China (e-mail: yanch@fudan.edu.cn; xzeng@fudan.edu.cn).

Sheng-Guo Wang is with the Department of Engineering Technology, University of North Carolina at Charlotte, Charlotte, NC 28223 USA.

Dian Zhou is with the State Key Laboratory of Integrated Chips and Systems, Microelectronics Department, Fudan University, Shanghai 200433, China, and also with the Department of Electrical Engineering, University of Texas at Dallas, Richardson, TX 75080 USA.

Digital Object Identifier 10.1109/TCAD.2024.3473829

Neural network-based thermal simulators attract widespread attention in recent years. Taking structural information and power-maps as input, they output temperature solutions at fixed points. These methods are based on autoencoders [11] and physics-informed neural networks [12], while lacking the configurations of 3-D power-maps [13]. An operator learning-based neural network thermal simulator [13] is proposed for 3D-ICs by learning a mapping from key configurations (including power-maps) in partial differential equations (PDEs) to temperature fields. However, the claimed 1 ms runtime [13] seems inadequate for online thermal prediction because of the extremely high invocation frequency. Meanwhile, these neural network methods require intricate network architectures, lack adequate physical interpretability as closed-boxes, and necessitate substantial time and computational resources for the training process.

The above methods obtain global solutions at one time. Nevertheless, in many real-world applications, such as thermal management, the focus is often solely on the temperatures of the hottest points, known as hot-spots. Consequently, those approaches are not sufficiently effective in obtaining the precise solutions for localized hot-spots.

On the contrary, based on the Feynman–Kac formula [14], stochastic methods can obtain local temperatures with high parallelism. Inspired by the application in matrix inversion [15], the random walk method on grids is applied for the thermal analysis of complex IC-interconnect systems [16] and large RC networks [17]. Besides, similar techniques are applied to floorplanning algorithms with considering the thermal via insertion [18] or temperature of 3D-ICs [19]. Nevertheless, these works only consider the Dirichlet BCs, i.e., isothermal surfaces, disregarding the Neumann and Robin BCs that are more realistic and prevalent in IC thermal analysis.

Neumann boundary is commonly treated as adiabatic boundary in engineering practice. Robin boundary, also known as convective boundary, corresponds to the interface that exchanges heat with the surrounding fluid, such as the surface of cooling fans and heat sinks. For thermal analysis with mixed BCs, the state-of-the-art random walk method [20] combines the walk-on-grids (WOGs) method and the walk-on-cuboids method. A widely used heuristic reflection technique [21] is utilized to handle the Neumann boundary. When dealing with the Robin boundary, a heuristic artificial treatment is adopted by constructing a virtual Dirichlet node outside the Robin boundary with the ambient temperature [20]. The walking particle can either be absorbed by the virtual Dirichlet node with a specific probability or be directly reflected by the Robin boundary. Nonetheless, when calculating temperatures at hot-spots with Robin BCs, [20] results in an unacceptable error of as much as 20 °C. As far as we are concerned, in the field of thermal analysis, there are currently no accurate random walk methods to well tackle Robin BCs.

In our preliminary work [22], we propose a novel parallelized path integral random walk (PIRW) solver based on the Feynman–Kac formula, the first accurate stochastic method for steady-state thermal analysis with Fourier heat transfer Robin BCs. Different from the commonly employed but inaccurate reflection treatment for Neumann BCs, our solver leverages rigorous computation of local time [23] to

evaluate the integral terms associated with Neumann and Robin BCs. More significantly, we introduce the Feynman–Kac functional $\hat{e}_c(t)$ [24] associated with Robin boundary into thermal analysis for the first time.

Thermal management focuses on controlling the real-time temperatures of running chips. Proactive methods predict future temperatures and take actions before thermal conflicts occur, thus effectively reducing thermal conflicts and trigger frequency. Accuracy of prediction algorithms, however, is critical in proactive thermal management, but they face the dilemmatic requirements of high accuracy and extremely high speed (low overhead). Although the original stochastic PIRW solver achieves more than 100× speedup over FEMs, a speed of 2.72 s [22] is far from sufficient for real-time predictions.

In this article, based on the high-precision PIRW solver, we further propose an ultrafast precalculation PIRW (pPIRW) predictor for proactive thermal management. Thanks to the fixed geometry and BCs of ICs in working scenarios, we modify the original PIRW solver by extracting the varying real-time power from the temperature expression, and precalculating the left structure-related terms, i.e., the attenuated Feynman–Kac functional $\hat{e}_c(t)$ of the Robin boundary in all paths, through massive random walks. This time consuming precalculation is offline, and the left online part enables ultrafast real-time temperature prediction through straightforward vector multiplication within microseconds.

Our contributions are summarized as follows.

- 1) The proposed PIRW solver innovatively adopts the accurate local time allocation and the Feynman–Kac functional $\hat{e}_c(t)$ of the Robin boundary in all path integrals to handle Neumann and Robin BCs with high precision, thereby becoming the first accurate stochastic method for steady-state thermal analysis with mixed BCs.
- 2) The high independence of paths and the similarity of particle behavior in each path make the PIRW solver naturally highly parallel. Leveraging the high parallelism, local accurate temperature solutions can be obtained at little time and space cost.
- 3) Separating varying power from other unchanging parts in the PIRW solver, we derive the pPIRW method, the steady-state thermal solution in closed form in (25). We apply pPIRW for proactive thermal management as an accurate and ultrafast temperature predictor with a clear physical explanation. With precalculating the offline part of attenuated $\hat{e}_c(t)$ in the heat-source region, temperatures can be obtained online by vector multiplication at the microsecond level.
- 4) After closely comparing the temperature expressions of the pPIRW and the CTM, we reveal that the pPIRW essentially calculates a partial inverse of the large-scale thermal resistance matrix $\mathbf{G}$ in the CTM by stochastic methods. This relationship has been numerically validated.
- 5) pPIRW is a discretization-free method with constant time complexity, without linear system construction and actual matrix inversion, making it particularly attractive for engineers working on large-scale or multiscale problems with complex geometries.

Experimental results show that the PIRW solver preserves high accuracy with an acceptable error within 0.5 °C,

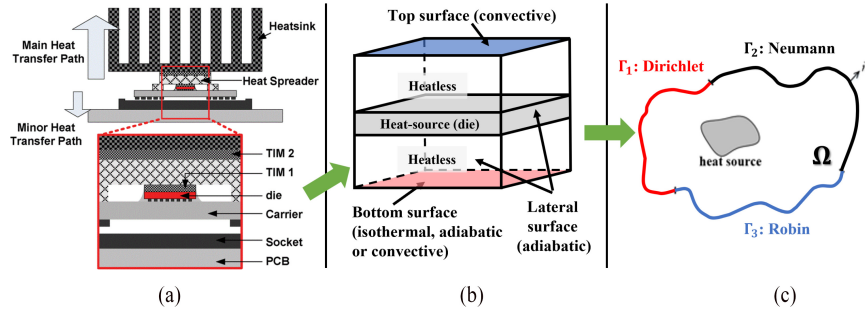


Fig. 1. Formulation of the steady-state thermal analysis. (a) Practical chip package [1]. (b) Simplified thermal model. (c) Mathematical model.

achieves $136\times\text{--}209\times$ speedup and $8.53\times\text{--}11.1\times$ storage space reduction with all three kinds of BCs, compared with the state-of-the-art CTM tool 3D-ICE. Furthermore, the pPIRW predictor achieves $2.5e4\times\text{--}6.7e6\times$ speedup and $36.6\times\text{--}42.5\times$ storage space reduction over PIrw without loss of accuracy. The thermal management with pPIRW can eliminate all heat conflicts and keep the highest operational frequency.

II. BACKGROUND

A. Steady-State Thermal Analysis Formulation

Fig. 1 illustrates the formulation of the steady-state thermal analysis problem, from the practical chip package [1] [Fig. 1(a)], to the simplified thermal model [Fig. 1(b)] and finally to the mathematical model [Fig. 1(c)].

From Fig. 1(a), the die (red color) within the chip package is the heat source for the temperature required. Other layers, such as the heat sink, heat spreader, thermal interface material (TIM), carrier, and PCB, are modeled as heatless regions.

Generally, the main heat transfer path is the heat sink (top surface) modeled as convective (Robin) boundary. The minor heat transfer path is the PCB (bottom surface) conditionally modeled as isothermal (Dirichlet), adiabatic (Neumann), or convective (Robin) boundaries. The four side surfaces, with negligible heat conduction, are usually modeled as adiabatic boundaries.

The mathematical model of 3-D steady-state heat conduction problem depicted in Fig. 1(c) can be formulated as a Poisson equation with mixed BCs

$$-\frac{1}{2}\nabla^2 T = f(x) \quad \text{in } \Omega \quad (1a)$$

$$T = \phi_1(x) \quad \text{on } \Gamma_1 \quad (1b)$$

$$\frac{\partial T}{\partial n} = \phi_2(x) \quad \text{on } \Gamma_2 \quad (1c)$$

$$\frac{\partial T}{\partial n} - cT = \phi_3(x) \quad \text{on } \Gamma_3 \quad (1d)$$

where T is the temperature in domain Ω , $(\partial T/\partial n)$ is outer normal derivative of T at boundary, and $\nabla^2(\cdot)$ is the Laplace operator. Γ_1 , Γ_2 , and Γ_3 are Dirichlet, Neumann, and Robin boundary with given BCs $\phi_1(x)$, $\phi_2(x)$, and $\phi_3(x)$, respectively. $f(x) = (p_d(x)/2k(x))$, where $p_d(x)$ is heat generation density, equal to 0 in heatless regions, and $k(x)$ is thermal conductivity. $c(x) = -(h(x)/k(x))$, where $h(x)$ is the convection coefficient at Robin boundary. If multiple materials are included, the temperature and heat flow at the interface are considered continuous.

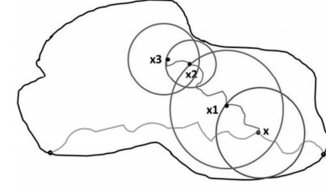


Fig. 2. WOSs method [25].

B. Walk-on-Spheres Method

Stochastic methods use massive repeated Brownian motion paths to approximate solutions of PDEs. For example, in a heatless domain with only Dirichlet boundary, the Poisson equation (1a) degenerates into a Laplace equation and the probabilistic solution is obtained as [26]

$$T(x) = E^x(\phi_1(X_{\tau_D})) = \frac{1}{N} \sum_{i=1}^N T(X_{\tau_D}) \quad (2)$$

where X_t is the Brownian motion and τ_D is the first time when each Brownian path hits Dirichlet boundary. i denotes a Brownian motion path and N is the total path number. E^x is the mathematical expectation.

To simulate each Brownian motion path, a floating random walk procedure in three dimensions can be formulated. In homogeneous regions, the temperature at the given point $T(x, y, z)$ can be described as

$$T(x, y, z) = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi T(r, \omega, \phi) \sin\phi d\phi d\omega \quad (3)$$

where r is the radius of the sphere centered at (x, y, z) , ω and ϕ are the polar and azimuth angles, and $T(r, \omega, \phi)$ is the boundary temperature at spherical surface. Therefore, a sampling-type walk-on-sphere (WOS) method employing spherical transition domains with variable radius can be constructed to calculate the temperature solution of each path.

During the WOS process, each step of a walk is from center to sphere surface, as illustrated in Fig. 2. For a single Brownian path, given a starting point x inside the domain D , one can draw a largest sphere centered at x and does not intersect with the domain boundary. Next, the second location x_1 is sampled on the surface of this sphere using a uniform distribution. Likewise, considering x_1 as the new starting point, another sphere with the maximum radius can be drawn and the third location x_2 can be obtained by uniform sampling. Repeat these operations until the Brownian path is finally absorbed and terminated by Dirichlet boundary at X_{τ_D} , and then we record

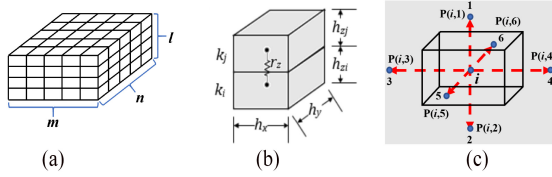


Fig. 3. WOGs method. (a) Grid discretization. (b) Thermal resistor between nodes [20]. (c) Transition probability in 3-D [$d(i) = 6$].

the temperature $T(X_{td})$ for accumulation. By simulating a large amount of the Brownian motion paths, the numerical solution of $T(x)$ in (2) can be obtained.

C. Walk-on-Grids Method

The WOG method [20] is widely utilized in the heat-source region to simulate the Brownian motion path. The heat-source region is evenly discretized into rectangular cuboid grid cells as shown in Fig. 3(a), and the grid needs to be fine enough to reasonably assume points inside each grid cell share the same temperature and power density. Particles jump among central nodes of these cuboid grid cells.

For adjacent nodes in Fig. 3(b), the thermal resistance r_z is

$$r_z = \frac{1}{2k_i} \cdot \frac{h_{zi}}{h_x h_y} + \frac{1}{2k_j} \cdot \frac{h_{zj}}{h_x h_y} \quad (4)$$

where k_i and k_j are the thermal conductivities of nodes i and j , h_{zi} and h_{zj} are the heights of the two adjacent grid cells, and h_x and h_y are the side lengths of the grid cell along the x and y -axis, respectively. r_x and r_y at node i can be calculated in the same way.

During the process, as shown in Fig. 3(c), the reward received at node i and the probability of a particle jumping from node i to adjacent node j can be presented as

$$r(i) = \frac{p(i)}{g_t(i)} \quad (5a)$$

$$P(i, j) = \frac{g(i, j)}{g_t(i)} \quad (5b)$$

where $g_t(i) := \sum_j^{d(i)} g(i, j)$, and $g(i, j)$ is the thermal conductance between two adjacent nodes i and j . $d(i)$ is the edge degree of node i . $p(i)$ is the power of the grid with central node i . The path will be terminated when the particle reaches Dirichlet boundary and receives the final reward, the corresponding Dirichlet boundary value. The average accumulated rewards obtained along the WOG path provide an estimation of the temperature at the starting point.

D. Proactive Thermal Management

Current thermal management methods roughly fall into two categories: 1) reactive and 2) proactive methods [27]. Reactive methods take approaches like dynamic voltage and frequency scaling (DVFS), task mapping, and task migration to reduce temperatures after detecting thermal conflicts from on-chip thermal sensors.

Instead, proactive methods anticipate future temperatures and take preemptive actions before thermal conflicts arise. When the system is running a set of applications, the management is triggered at regular intervals, predicting future

temperatures at chip hot-spots to decide if measures are needed to reduce temperatures.

The proactive management process can be formulated as an optimization problem [28]

$$\begin{aligned} &\text{maximize} \quad \text{average } V/f \text{ level} \\ &\text{s.t.} \quad T \leq T_{\text{threshold}} \end{aligned} \quad (6)$$

where the object is to maximize the speed, i.e., average voltage or frequency (V/f) of cores when executing a set of applications, while keeping temperature T under the threshold $T_{\text{threshold}}$.

The effectiveness of proactive thermal management is highly dependent on the accurate temperature predictor inside, which also needs to be ultrafast and lightweight for frequent online use.

Currently, temperature predictions generally rely on data-driven empirical models, such as the autoregressive moving average model [29], the neural network [30], and the regression model (RM) [31], or physical models, such as CTMs.

The empirical predictor with RM has been proposed [31] as

$$\tilde{T}_t = \beta_0 + \beta_1 T_{t-1} + \beta_2 T_{t-2} + \beta_3 P_t + \beta_4 P_{t-1}. \quad (7)$$

Fundamentally, RM proposes a linear closed-form expression between the predictive temperature T_t , and the previous temperatures T_{t-1} and T_{t-2} , and the previous and current power P_{t-1} and P_t . All other coefficients β_0 – β_4 are fitted from actual data obtained from power and temperature sensors.

For steady-state thermal analysis, CTM as a FDM, discretizes the whole domain, and constructs linear systems as

$$\mathbf{GT} = \mathbf{P} + T_{\text{amb}} \mathbf{g}_{\text{amb}} \quad (8)$$

where $\mathbf{T}$ is the required temperature, $\mathbf{G}$ is the thermal conductance matrix, T_{amb} is the ambient temperature, $\mathbf{g}_{\text{amb}}$ is the thermal conductance at boundary, and $\mathbf{P}$ is the power-map.

The CTM predictor [5] inverts the thermal conductance matrix $\mathbf{G}$ directly and gives the temperature $\mathbf{T}$ a explicit closed form as

$$\mathbf{T} = \mathbf{G}^{-1} \mathbf{P} + T_{\text{amb}} \mathbf{G}^{-1} \mathbf{g}_{\text{amb}}. \quad (9)$$

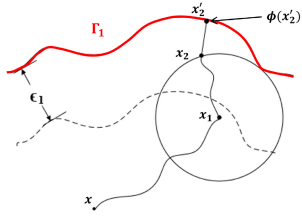
Therefore, the time complexity of the CTM predictor in the offline phase is $O(n^3)$, where n is degrees of freedom.

III. PROPOSED HIGH-PRECISION STOCHASTIC THERMAL SOLVER

This section introduces the probabilistic solution to the Poisson equation with mixed BCs and demonstrates the innovative techniques implemented in our PIRW solver.

A. Probabilistic Solution for Poisson Equation With Mixed Boundary Conditions

The Feynman–Kac formula [14] establishes a link between PDEs and stochastic processes, which can be approximated by the Monte Carlo method with massive repeated standard reflecting the Brownian motion (SRBM) path simulations. Given the Poisson equation (1a) in domain Ω with mixed

Fig. 4. Absorption in ϵ_1 -region near Dirichlet boundary.

BCs (1b)–(1d), the Feynman–Kac formula gives the probabilistic solution as [14]

$$T(x) = E^x \left\{ \hat{e}_c(\tau_{\Gamma_1}) \phi_1(X_{\tau_{\Gamma_1}}) \right\} \quad (10a)$$

$$+ E^x \left\{ \int_0^\infty \hat{e}_c(t) \phi_{2,3}(X_t) dL(t) \right\} \quad (10b)$$

$$+ E^x \left\{ \int_0^\infty \hat{e}_c(t) f(X_t) dt \right\} \quad (10c)$$

$$\hat{e}_c(t) := e^{\int_0^t c(X_s) dL(s)} \quad (10d)$$

where X_t is the SRBM and $\hat{e}_c(t)$ is the Feynman–Kac functional defined on Robin boundary [24]. τ_{Γ_1} is the time when the Brownian path first hits Dirichlet boundary. $L(t)$ is the corresponding local time near Neumann and Robin boundaries [23]. The integral terms in (10a)–(10c) represent contributions of Dirichlet BCs, Neumann and Robin BCs, and the right-hand side (r.h.s.) of the Poisson equation, respectively.

The Skorokhod equation defines SRBM as [23], [32]

$$X_t = B_t - \int_0^t n(X_s) L(ds) \quad (11)$$

where $n(x)$ denotes the outward normal at $x \in \partial\Omega$. If B_t is a standard Brownian motion (SBM), the corresponding X_t will be an SRBM and the local time $L(t)$ increases only when $X_t \in \Gamma_2 \cup \Gamma_3$. Intuitively, the path of SRBM behaves like the SBM in the interior of the domain and reflects inward along the normal direction when reaching Neumann or Robin boundary.

B. Dirichlet Boundary Conditions

For Dirichlet BCs, the discrete form of (10a) can be expressed as

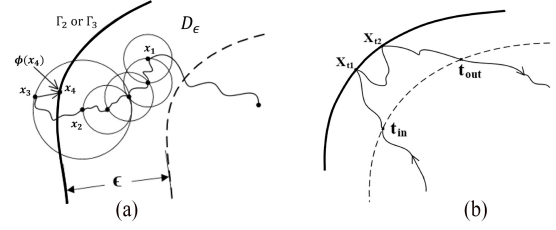
$$\tilde{T}_1(x) = \frac{1}{N} \sum_{i=1}^N \left\{ \hat{e}_c(\tau_{\Gamma_1}) \phi_1(X_{\tau_{\Gamma_1}}) \right\} \quad (12)$$

where i denotes the i th SRBM path of total N paths and $\phi_1(X_{\tau_{\Gamma_1}})$ is the Dirichlet boundary value at $X_{\tau_{\Gamma_1}}$. $\hat{e}_c(\tau_{\Gamma_1})$ is contributed by Robin BCs.

Generally, a thin absorption strip with width ϵ_1 is constructed near Dirichlet boundary. As shown in Fig. 4, if the particle jumps to x_2 inside the ϵ_1 -region via x and x_1 by WOS, it will be absorbed by Dirichlet boundary Γ_1 , and the temperature $\phi_1(x'_2)$ will be recorded, where x'_2 is the nearest point to x_2 on Γ_1 .

C. Neumann Boundary Conditions

The Neumann boundary is often empirically addressed as adiabatic boundary with $\phi_2(x) = 0$ in engineering, and

Fig. 5. Techniques to handle the reflecting BCs in PIrw. (a) Particle's behavior ϵ -region [25]. (b) Calculation of the local time.

can be roughly handled with traditional path reflection technique [20], [21]. In this article, a more precise method with computation of local time is utilized to deal with Neumann BCs [25], which adheres strictly to correct mathematical formulation.

The local time $L(t)$ can be expressed as [23]

$$L(t) \equiv \lim_{\epsilon \rightarrow 0} \frac{\int_0^t I_{D_\epsilon}(X_s) ds}{2\epsilon} \quad (13)$$

where D_ϵ is a strip region in the domain containing $\partial\Omega$ near Γ_2 , and ϵ is the strip region width. I is the indicator function. $L(t)$ can be seen as a measure of the time that the SRBM spends near the Neumann boundary. For Neumann BCs, the discrete form of (10b) can be written as

$$\tilde{T}_2(x) = \frac{1}{N} \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_j) \phi_2(X_{t_j}) dL(t_j) \right\} \quad (14)$$

where j' denotes each step of the path and j denotes the steps when the path hits Neumann boundary. NP is the maximum number of path steps. Every time the path is reflected, the value of the term $\sigma T(X_{t_j}) = \hat{e}_c(t_j) \phi_2(X_{t_j}) dL(t_j)$ at the collision point X_{t_j} is accumulated.

As illustrated in Fig. 5(a), inside the D_ϵ region, the particle executes a fixed radius WOS scheme with $r = \Delta x$ or $2\Delta x$. Assuming the particle first enters D_ϵ at x_1 , it can subsequently execute WOS with a fixed radius of Δx . When the particle jumps to a location, such as x_2 , the distance of which to the Neumann boundary is less than Δx , the WOS radius is adjusted to $2\Delta x$, and the particle possibly jumps out of the domain to x_3 . Subsequently, we pull it back to the nearest location x_4 on the boundary, record the corresponding term $\sigma T(x_4)$, and resume the WOS process starting from x_4 . More details and further explanations can be found in [25].

Within D_ϵ , the corresponding elapsed time of one step for local time could be $(\Delta x)^2/3$ or $(2\Delta x)^2/3$, depending on the jump radius. Hence, adopting an accurate allocation scheme [33], the local time increment $dL(t_j)$ corresponding to the collision point X_{t_j} can be approximated as

$$dL(t_j) \approx \delta n_{t_j} \frac{(\Delta x)^2}{6\epsilon}, \quad \delta n_{t_j} = \begin{cases} \frac{n_{j,\text{out}} - n_{j,\text{in}}}{m_j}, & \text{if } m_j > 0 \\ 0, & \text{if } m_j = 0 \end{cases} \quad (15)$$

where $n_{j,\text{out}} - n_{j,\text{in}}$ represents the coefficient increment of the local time between entering and exiting D_ϵ , as shown in Fig. 5(b). m_j is the number of collisions at the boundary during the same period between t_{in} and t_{out} .

D. Robin Boundary Conditions

Similar to Neumann boundary, the local time increment (15) and the corresponding WOS scheme in D_ϵ are adopted by Robin boundary. For Robin BCs, the discrete form of (10b) can be calculated as

$$\tilde{T}_3(x) = \frac{1}{N} \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_j) \phi_3(X_{t_j}) \delta n_{t_j} \frac{(\Delta x)^2}{6\epsilon} \right\} \quad (16)$$

where j denotes the steps where the path hits Robin boundary. The term $\sigma T(X_{t_j})$ of each collision point X_{t_j} at Robin boundary is counted in the cumulative sum.

The discrete form of the Feynman–Kac functional (10d) can be written as

$$\hat{e}_c(t_j) = e^{\sum_{k=0}^j c(X_{t_k}) \delta n_k \frac{(\Delta x)^2}{6\epsilon}}. \quad (17)$$

We notice that (17) is only modified when the path hits Robin boundary, since it contains a negative coefficient c that only appears in Robin BCs (1d). When the path hits the Robin boundary, the modification in (17) is manifested in an abrupt reduction of the exponential term of $\hat{e}_c(t)$. As the number of hitting Robin boundary increases, $\hat{e}_c(t)$ gradually decays from 1 to 0. Note that, all integral terms in (10a) contain the functional $\hat{e}_c(t)$. Therefore, the value of $\hat{e}_c(t)$ requires to be recorded and leveraged during the entire SRBM path, and its attenuation leads to the final convergence of the whole path integral (10a).

Particularly, when Robin boundary is excluded in the mixed boundary problem, $\hat{e}_c(t) \equiv 1$, meaning that Dirichlet and Neumann boundary can be handled by traditional absorbing and reflecting random walk methods. Hence, these conventional stochastic methods with consideration of boundary are just special cases of our proposed method.

E. Heat Source: Combination of WOS and WOG

Due to the high efficiency of WOS and the effectiveness of WOG for issues with internal heat generation, we utilize WOS in heatless regions and WOG in heat-source regions. Thus, the discrete form of (10c) in the heat-source region can be calculated as

$$\tilde{T}_r(x) = \frac{1}{N} \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_p) r(X_{t_p}) \right\} \quad (18)$$

where p denotes the steps in the heat-source region. $r(X_{t_p})$ is the reward obtained at each node X_{t_p} the path goes through, as defined in (5a). Each time the particle jumps within the heat-source region, the value of the term $\hat{e}_c(t_p) r(X_{t_p})$ at each node X_{t_p} is accumulated.

We construct a virtual grid layer in the heatless area (white) close to the heat-source region (grey), as illustrated in Fig. 6. When the particle enters into the virtual layer at x_2 through x_1 by WOS, we pull it to the center x'_2 of the corresponding grid cell. Then, with assigning power $p(i) = 0$ to all nodes in the virtual layer, the WOG procedure is adopted and the particle makes a random jump to an adjacent node.

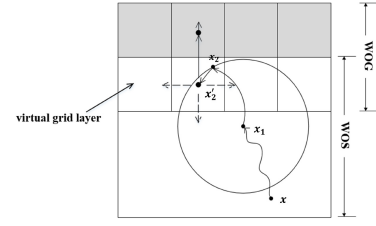


Fig. 6. Switch between WOS and WOG at the interface.

Algorithm 1 PIRW Solver for Steady-State Thermal Analysis With Mixed BCs

Input: Geometry and thermal conductivity of domain, boundary conditions, path number N , maximum step number NP of each path, point x_0 .

Output: Temperature at point $\tilde{T}(x_0)$.

- 1: *Initialization:* For the i th path, path integral $\tilde{T}_i(x_0) = 0$, Feynman-Kac functional $\hat{e}_c^{(i)}(t) = 1$;
- 2: **for** $i = 1 : N$ path **do**
- 3: $X = x_0$;
- 4: **for** $j' = 1 : NP$ step **do**
- 5: **if** X in heat-source region **then**
- 6: Update $\tilde{T}_i(x_0)$ with (18);
- 7: WOG as Sec. III-E;
- 8: **else if** X on Dirichlet boundary **then**
- 9: Update $\tilde{T}_i(x_0)$ with (12);
- 10: Be absorbed at X and exit current path;
- 11: **else if** X on Neumann boundary **then**
- 12: **for** collision X_{t_j} **do**
- 13: Update $\tilde{T}_i(x_0)$ with (14);
- 14: WOS with fixed $R = \Delta x$ or $2\Delta x$ as Sec. III-C;
- 15: **else if** X on Robin boundary **then**
- 16: **for** collision X_{t_j} **do**
- 17: Update $\hat{e}_c^{(i)}(t)$ with (17);
- 18: Update $\tilde{T}_i(x_0)$ with (16);
- 19: WOS with fixed $R = \Delta x$ or $2\Delta x$ as Sec. III-C;
- 20: **else**
- 21: WOS as Sec. II-B;
- 22: **Return** $\tilde{T}(x_0) = \frac{1}{N} \sum_{i=1}^N \tilde{T}_i(x_0)$;

F. Framework of the PIRW Solver

In conclusion, the discrete form of the probabilistic solution (10a) of the Feynman–Kac formula [14] is

$$\tilde{T}(x) = \tilde{T}_1(x) + \tilde{T}_2(x) + \tilde{T}_3(x) + \tilde{T}_r(x) \quad (19)$$

where the four right-hand side terms correspond to temperature contributions of Dirichlet boundary, Neumann boundary, Robin boundary, and the heat-source region, respectively.

The framework of the proposed PIRW solver is summarized in Algorithm 1. At each step in a path, the particle position needs to be determined first. If the particle is in the heat-source region, we calculate (18) and utilize the WOG method. When it is on Dirichlet boundary, (12) can be calculated and the path is terminated. If the particle is on Neumann boundary, we compute (14) and adopt the improved WOS method as Section III-C. Similar operations are performed if it is on Robin boundary, together with updating the Feynman–Kac

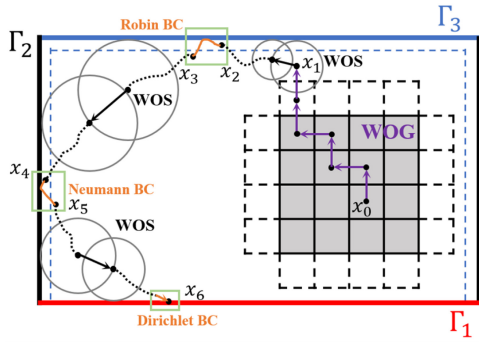


Fig. 7. Illustration of the overall walking strategy.

functional $\hat{e}_c(t)$ as Section III-D. Otherwise, a trivial WOS method is utilized.

Fig. 7 intuitively illustrates a typical random walk trajectory. The path starts at temperature-required x_0 in the heat-source region, performs WOG until it exits the heat-source region at x_1 , and then switches to WOS in the heatless region. If it nears Neumann or Robin boundaries (x_2 and x_4), WOS is performed with a fixed radius as described in Section III-C. The path finally terminates if it reaches the Dirichlet boundary at x_6 or the threshold NP , the maximum number of hops.

The novelty of the proposed PIrw solver can be concluded as follows.

- 1) Unlike the traditional heuristic “reflection-absorption” treatment for Robin BCs, PIrw is the first stochastic method for steady-state thermal analysis with Robin BCs, based on the Feynman–Kac formula [14] and the mathematically correct calculation of local time [33].
- 2) PIrw significantly improves the accuracy of stochastic methods for thermal analysis problems involving the convective boundary, reducing the error from 7 °C to 0.5 °C.

IV. PROPOSED ULTRAFAST PRECALCULATION STOCHASTIC THERMAL PREDICTOR

Generally, an operating chip has fixed geometry and BCs, i.e., fixed package and cooling methods, and the only varying part is the real-time changed power. If the varying power can be separated from the stochastic temperature expression (19) and other fixed terms can be calculated offline, an ultrafast temperature prediction can be achieved. In this section, we further propose a precalculation stochastic thermal predictor, based on our proposed stochastic solver PIrw, with varying right hand, i.e., power stimulates.

A. Principle of pPIRW Predictor

Substituting (5a) into (18), the discrete probabilistic solution (19) can be written as

$$\tilde{T}(x) = \tilde{T}_b(x) + \frac{1}{N} \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_p) \frac{p(X_{t_p})}{g_t(X_{t_p})} \right\} \quad (20)$$

where $\tilde{T}_b(x) = \tilde{T}_1(x) + \tilde{T}_2(x) + \tilde{T}_3(x)$ is the sum of the temperature contributions of all three kinds of boundary. Fortunately, $\tilde{T}_b(x)$ is only related to the domain geometry

and BCs, irrelevant to the heat source. Therefore, it can be calculated in advance.

During the WOG process inside the heat sources, since a particle only jumps to central nodes of finite grid cells, $p(X_{t_p})$ and $g_t(X_{t_p})$ only take a finite number of constant values. For the heat-source region with $N_g = m \times n \times l$ cuboid grid cells shown in Fig. 3(a), (20) can be rewritten as

$$\tilde{T}(x) = \tilde{T}_b(x) + \frac{1}{N} \sum_{k=1}^{N_g} \left\{ \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_{p_k}) \frac{p(X_{t_{p_k}})}{g_t(X_{t_{p_k}})} \right\} \right\} \quad (21)$$

where p_k denotes the steps at the center of the k th grid cell in the heat-source region, and $X_{t_{p_k}}$ is the central node location at time t_{p_k} .

Independent with the paths, both power $p(X_{t_{p_k}})$ and thermal conductance $g_t(X_{t_{p_k}})$ are only functions of positions in the heat-source region. Therefore, using the cell number k to indicate the position $X_{t_{p_k}}$, (21) becomes

$$\tilde{T}(x) = \tilde{T}_b(x) + \frac{1}{N} \sum_{k=1}^{N_g} \left\{ \frac{p(k)}{g_t(k)} \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_{p_k}) \right\} \right\} \quad (22)$$

where $\hat{e}_c(t_{p_k})$, however, is still path-dependent, and cannot be moved out of the double loop of i and j' .

Considering only changing part is power $p(k)$ in thermal management, (22) can be further rewritten as

$$\tilde{T}(x) = \tilde{T}_b(x) + \sum_{k=1}^{N_g} \{p(k) \cdot a(k)\} \quad (23)$$

where

$$a(k) := \frac{1}{N} \frac{\hat{e}_{\text{sum}}(k)}{g_t(k)}, \quad \hat{e}_{\text{sum}}(k) := \sum_{i=1}^N \left\{ \sum_{j'=0}^{NP} \hat{e}_c(t_{p_k}) \right\}. \quad (24)$$

In vector form, (23) becomes

$$\tilde{T}(x) = \mathbf{a}^T \mathbf{p} + \tilde{T}_b(x) \quad (25)$$

where the column vectors $\mathbf{a} = [a(k)]_{N_g \times 1}$ and $\mathbf{p} = [p(k)]_{N_g \times 1}$, $k = 1, \dots, N_g$. Given the precalculated $\tilde{T}_b(x)$, $\mathbf{a}$ and the varying power $\mathbf{p}$, the random walk temperature solution of the start point x can be calculated efficiently from (25) by simple vector multiplication and addition.

If temperatures of multiple points x_i ($i = 1, \dots, k'$) are interested, the matrix form of (25) is written as

$$\tilde{\mathbf{T}}(\mathbf{x}) = \mathbf{A}^T \mathbf{p} + \tilde{\mathbf{T}}_b(\mathbf{x}) \quad (26)$$

where $\tilde{\mathbf{T}}(\mathbf{x}) = [\tilde{T}(x_i)]_{k' \times 1}$, $\tilde{\mathbf{T}}_b(\mathbf{x}) = [\tilde{T}_b(x_i)]_{k' \times 1}$, and $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{k'}]_{N_g \times k'}$. Consequently, all $\tilde{T}_b(x_i)$ and $\mathbf{a}_i$ are required to be precalculated, respectively. With precalculation terms, only simple vector calculation (25) or the matrix calculation (26) is required to be performed. Considering the fixed geometry and BCs of working chips, we refer to (25) or (26) as the online part of pPIRW predictor.

The memory usage and speed of the online part are related to the power vector dimension N_g , which depends on the volume discretization of heat-source regions, and can be reduced according to actual power distribution.

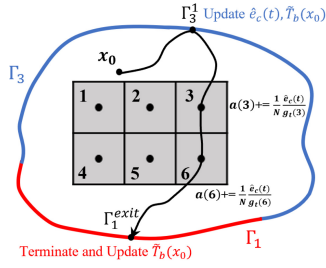


Fig. 8. Illustration of a path of the pPIRW.

B. Framework of pPIRW Predictor

Fig. 8 shows a specific path as an example, starting from x_0 , hitting Robin boundary point Γ_3^1 , traversing the nodes 3 and 6 in the source domain, and finally absorbed by the Dirichlet boundary at Γ_1^{exit} , respectively. When hitting Robin boundary point Γ_3^1 , the current $\hat{e}_c(t)$ and $\tilde{T}_b(x_0)$ are updated. When traversing nodes 3 and 6 in the source domain, $a(3)$ and $a(6)$ are accumulated. When the path terminates at the absorbing point Γ_1^{exit} , $\tilde{T}_b(x_0)$ is updated.

The physical meaning of this path is that the heat generated from nodes 3 and 6 is taken away by the Robin boundary Γ_3^1 . Thus, $a(k)$ is the probability of the heat of the k -th grid diffusing to all Robin boundaries. In fact, paths are solely determined by the geometry, more precisely by the Green's function of the domain.

Mathematically, from (25), the temperature is dependent on the geometry domain $\mathbf{a}$, BCs $\tilde{T}_b$, and the power-map $\mathbf{p}$. $\mathbf{a}$ is dependent only on the Green's function of the domain. $\tilde{T}_b$ is on the fixed Dirichlet, Neumann, and Robin BCs. In thermal management scenarios, $\mathbf{a}$ and $\tilde{T}_b$ are fixed, and the only change being the power-map $\mathbf{p}$. Therefore, the precise and ultrafast pPIRW predictor can be derived.

The precalculation part of pPIRW is devoted to obtaining values of $\mathbf{a}$ and $\tilde{T}_b$, similar to PIRW, but we need to temporarily record the full information of a path, and maintain vector $\mathbf{a}$, $\tilde{T}_b(x)$, $dL(t_j)$, and $\hat{e}_c(t)$ of the current step. The detailed process of precalculation part for a point is illustrated in Algorithm 2.

When the particle is in the k -th grid of the heat-source region, accumulate $\hat{e}_c(t_{p_k})/g_i(k)$ to $\mathbf{a}(k)$ as shown in the sixth line and take the WOG approach. Otherwise, when the particle is on boundary or in the heatless region, a treatment similar to Algorithm 1 is adopted, except that the accumulated temperature variable is changed from the total temperature $\tilde{T}_i(x_0)$ to the boundary temperature contribution $\tilde{T}_b^{(i)}(x_0)$.

The accuracy and speed of random walk methods depend on the path number N and the maximum step number NP in a path. Intuitively, path number N should be large enough to guarantee precision. As unbiased Monte Carlo solvers, the variance of PIRW and pPIRW is inversely proportional to path number N , according to the central limit theorem [20]. Meanwhile, the maximum step number NP should be large enough to ensure the convergence of path integrals (19), especially for the cases only Neumann and Robin BCs involved. Hence, a tiny tolerance ε is given for decaying $\hat{e}_c(t)$, and a path is terminated when $\hat{e}_c(t_j) \leq \varepsilon$.

Algorithm 2 Framework of pPIRW Thermal Predictor

Input: Domain Ω , boundary conditions, path number N , maximum step number NP of each path, tolerance ε for decaying $\hat{e}_c(t)$, heat-source region grids, point x_0 .

Output: Vector $\mathbf{a}$ and boundary contribution $\tilde{T}_b(x_0)$.

- 1: **Initialization:** For the i -th path, $\tilde{T}_b^{(i)}(x_0) = 0$, $\hat{e}_c^{(i)}(t) = 1$; $\mathbf{a} = \mathbf{0}$;
- 2: **for** path $i = 1 : N$ **do**
- 3: $X = x_0$;
- 4: **while** $\hat{e}_c(t_j) > \varepsilon$ **do**
- 5: **if** X in heat-source region of k -th grid **then**
- 6: $\mathbf{a}(k) += \hat{e}_c^{(i)}(t_{p_k})/g_i(k)$;
- 7: WOG;
- 8: **else**
- 9: Replacing $\tilde{T}_i(x_0)$ with $\tilde{T}_b^{(i)}(x_0)$, execute the same operation as line 8 to line 21 of Algorithm 1;
- 10: **Return** $\mathbf{a} = \mathbf{a}/N$ and $\tilde{T}_b(x_0) = \frac{1}{N} \sum_{i=1}^N \tilde{T}_b^{(i)}(x_0)$;

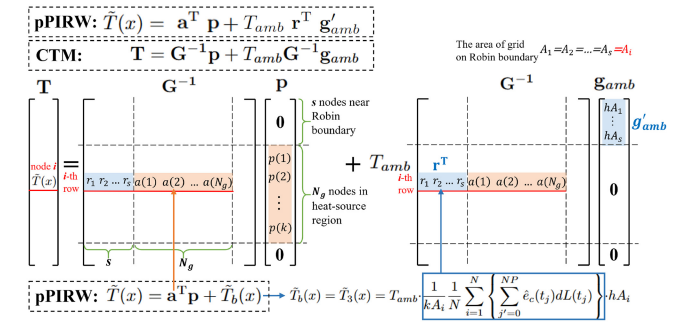


Fig. 9. Relationship between CTM and pPIRW predictors.

C. Relationship Between CTM and pPIRW

Comparing (26) and (9), we find the expressions of temperature by the pPIRW and the CTM are very similar indeed as shown in Fig. 9, and we will reveal their consistency carefully in this section.

The CTM is usually applied for a special but practical scenario without Dirichlet boundary, and includes only adiabatic Neumann boundary ($\partial T/\partial n = 0$) and Robin boundary, i.e., only the Robin BC contributions are included. Therefore, for a special case with only Robin BCs, temperature $\tilde{T}_b(x)$ in pPIRW can be simplified as $\tilde{T}_b(x) = \tilde{T}_1(x) + \tilde{T}_2(x) + \tilde{T}_3(x) = \tilde{T}_3(x)$.

In CTM, the difference matrix $\mathbf{G}$ is constructed and the inverse matrix $\mathbf{G}^{-1}$ is fully calculated offline. However, if only the temperature at site x , i.e., $T(x)$ is interested, the stochastic pPIRW as a local method partially calculates a row ($\mathbf{a}^T$ and $\mathbf{r}^T$) of the inverse matrix $\mathbf{G}^{-1}$ from CTM. The similar idea of “matrix inversion partially by a Monte Carlo method” can be found in [15].

The consistency between pPIRW and CTM can be validated through numerical experiments. We calculate the difference between $(\mathbf{r}^T, \mathbf{a}^T)$ obtained from the pPIRW and the corresponding row of the inverse matrix $\mathbf{G}^{-1}$ from the CTM. The relative L2 norm error of $\mathbf{a}$ is less than 2.34%, and the relative sum errors of $\mathbf{a}$ and $\mathbf{r}$ are only 0.18% and 0.0006%, respectively. More experimental details can be found

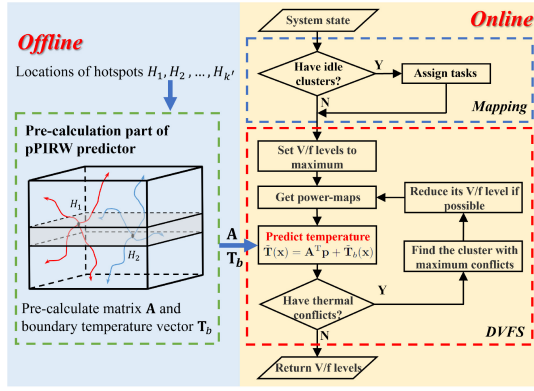


Fig. 10. Flow of proposed thermal management framework.

in Section V-D. Therefore, the online part of two temperature predictions, the CTM and the pPIRW, are basically the same with similar accuracy and speed.

Meanwhile, as a mesh-free method, pPIRW does not construct the matrix $\mathbf{G}$ itself, which is usually formed by laborious global discretization and the matrix assembly in CTM or other deterministic methods. Besides, pPIRW does not actually calculate the full inversion of the large matrix $\mathbf{G}$. Therefore, pPIRW saves much CPU time and memory resources in the offline part. As shown in experimental results, pPIRW achieves $29.2\times$ speedup and $634\times$ storage reduction over CTM under the same accuracy.

Furthermore, other minor merits, perhaps attractive for engineering applications, include accurate and unified handling of all three BC types without special treatments, natural parallelism, and simple source code.

D. Flow of Proactive Thermal Management With pPIRW Predictor

In this section, we put forward a proactive thermal management flow with pPIRW involved for multicore processors, as shown in Fig. 10.

In the offline phase, pPIRW executes random walks to calculate $\mathbf{A}$ and boundary temperature vector $\hat{\mathbf{T}}_b$ in (26). In the online phase, a proactive thermal management flow is presented. Our strategy combines task mapping and DVFS at cluster level. Given a set of tasks, the objective is to schedule the processor to efficiently complete all tasks and adopt the *optimal* thermal management strategy, where *optimal* means maximizing the efficiency (average V/f level), while keeping temperatures below the given threshold.

In *mapping* stage, we dequeue tasks with the same number of free clusters, sort them in descending order of average powers, and sequentially allocate these tasks to free clusters far away from the hottest clusters and with the lowest average temperatures.

In *DVFS* adjustment stage, we iteratively adjust V/f levels of all clusters from the highest voltage and frequency to the lowest, until there are no thermal conflicts or all clusters are running at the lowest V/f level. In each iteration, after determining core powers of tasks at the corresponding V/f level, we invoke the online part of pPIRW predictor to obtain hot-spot temperatures on all cores, estimate the number of

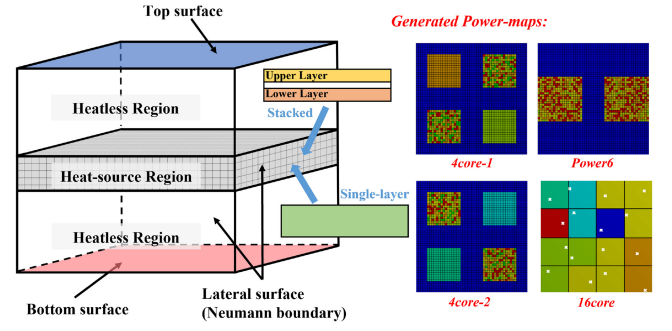


Fig. 11. Structure of chip thermal model and generated power-maps.

thermal conflicts of each cluster, and downgrade the V/f level of the cluster to minimize thermal conflicts.

V. EXPERIMENTAL RESULTS

To validate the accuracy and efficiency of the proposed PIRW and pPIRW solvers, we compare our solvers with the random method HRW [20], widely used FEM software ANSYS [2], and the state-of-the-art CTM tools HotSpot and 3D-ICE in thermal simulation for nanoscale CMOS technologies. To demonstrate the applicability of the pPIRW predictor in thermal management, we also compare our pPIRW predictor with the empirical RM predictor [31] and the CTM predictor under different thermal scenarios. Involved solvers and thermal management methods are implemented in C++. All experiments are carried out on a Linux workstation with Intel Xeon 6230N CPUs of 2.30 GHz frequency and 376 GB Memory.

A. Experimental Setting

In thermal analysis, the thermal model of the packaging chip can be divided into three parts, as shown in Fig. 11. The middle gray area is the discretized heat-source region, modeling transistor, and metal interconnection layers. The top cuboid is the heat transfer medium and the bottom cuboid is the substrate, both of which are heatless regions. Lateral surfaces are fixed as adiabatic Neumann boundary ($\partial T/\partial n = 0$). The top surface is set as Dirichlet or Robin boundary, and the bottom surface is regarded as Dirichlet, Robin, or adiabatic Neumann boundary. Both the top and bottom heatless cuboid regions measure $2\text{ cm} \times 2\text{ cm}$, with a thickness of $500\text{ }\mu\text{m}$ and a thermal conductivity of $395\text{ W/(K}\cdot\text{m)}$. The heat-source cuboid region has the same size, with a thickness of $100\text{ }\mu\text{m}$ and a thermal conductivity of $125\text{ W/(K}\cdot\text{m)}$. The ambient temperature is $20\text{ }^\circ\text{C}$.

As shown in Fig. 11, we generate power-maps for different chips and test them with different BCs. *4core-1* and *4core-2* are randomly generated power-maps, designed for a four-core chip. *Power6* is a random map for the POWER6 microprocessor with the same size, similar to structure 2 in [20]. *16core* is a power-map generated from real-time simulations in Section V-E for a 16-core chip. *4core-1* + *4core-2* is a two-layer stacked power-maps for cases with volumetric heat sources.

Table I lists nine cases with different BCs on the top and bottom surfaces, where D, R, and AN represent the Dirichlet boundary with $70\text{ }^\circ\text{C}$, the Robin boundary with thermal

TABLE I
CASE SETTING

Case	Top Surface	Bottom Surface	Boundary Type	Powermap
D4	D	D	1,2	4core-1
R4	R	R	2,3	
Mix4	R	D	1,2,3	
DP	D	D	1,2	Power6
RP	R	R	2,3	
Stack-1	R	D	1,2,3	4core-1(lower)+ 4core-2(higher)
Stack-2	R	R	2,3	
TM-1	R (12500)	AN	2,3	16core
TM-2	R (10000)			

convective coefficient $h = 8700 \text{ W/(K}\cdot\text{m}^2)$ by default, and the adiabatic Neumann boundary, respectively.

The very fine-grained discrete ANSYS [2] results (ANSYS_H with the element size of $6\text{e-}5 \text{ m}$) are utilized as the golden reference. The mesh method in ANSYS is chosen to be tetrahedrons. Since the original HotSpot and 3D-ICE do not support Dirichlet boundary directly, we approximate it by setting a very large thermal convective coefficient $h = 1\text{e}12 \text{ W/(K}\cdot\text{m}^2)$ on Robin boundary. In both stochastic methods, the strip width ϵ_1 of Dirichlet boundary is $1\text{e-}8\text{m}$, and the WOS radius Δx near boundary is $5\text{e-}7\text{m}$. The strip width ϵ of Neumann and Robin boundary is $1.36\Delta x$. The number of paths N is set to $5\text{e}3$, and adjusted to $1.5\text{e}3$ in the cases without Dirichlet boundary. The maximum step number NP of each path is $6\text{e}6$ in PIRW, and we set the tolerance ε for path termination as $5\text{e-}4$ in pPIRW.

B. Accuracy and Speed Comparison

1) *Accuracy Comparison of Traditional Stochastic Method and PIRW for Robin BCs:* We compare the accuracy of PIRW and traditional random method HRW in four cases to evaluate their ability in handling Robin BCs, as shown in Table II. With only Dirichlet BCs, i.e., case *D4* and *DP*, the maximal absolute error of HRW is 0.21°C and our proposed PIRW is 0.075°C . Both results are accurate enough for engineering applications. However, once Robin BCs are involved, such as case *R4* and *RP*, the absolute error of HRW becomes 6.917°C , unacceptable for engineering thermal analysis. In contrast, our PIRW possesses fairly high precision with absolute error within 0.5°C . Note that the maximal absolute error of HRW may even increase to 20°C [20] for cases with Robin BCs. These cases offer compelling evidence of the enhanced accuracy achieved by the proposed PIRW method in thermal analysis with Robin BCs.

Admittedly, the precise results of PIRW come at the cost of longer CPU time. To ensure convergence, particles of our PIRW reflect continuously on Robin boundary, as required by the rigorous calculation of local time. Particles of HRW, instead, can be absorbed directly on Robin boundary with a given probability, so their walking paths are terminated quickly. Objectively, if the temperature solver is not accurate enough, the speed advantage may not make much sense. Therefore, HRW is not suitable for thermal analysis due to intolerable errors.

2) *Accuracy and Speed Comparison of Stochastic Methods and FEMs With Mixed BCs:* To verify PIRW and pPIRW

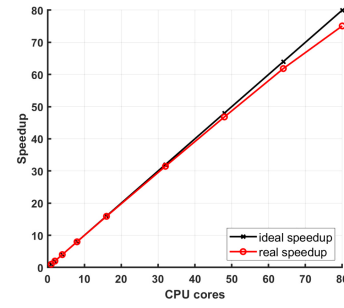


Fig. 12. Scalability of parallel PIRW.

can effectively and accurately handle all three types of BCs simultaneously, another four cases are evaluated. In cases *Mix4* and *Stack-1*, the chip model incorporates all three types of BCs. To fairly compare the speed of stochastic solvers and FEM methods, we construct the medium precision FEM, ANSYS_M in Table III, with similar accuracy to PIRW.

In Table III, we compare the results obtained by ANSYS, HotSpot, 3D-ICE, PIRW, and pPIRW under the same settings. Experimental results show that both stochastic methods PIRW and pPIRW can achieve high precision with maximal absolute error within 0.152°C in all cases. 3D-ICE offers similar high accuracy with maximum absolute error 0.287°C . HotSpot's maximum error is 2.184°C , consistent with its reported "usually less than 3%" in [7].

In all four cases, 3D-ICE shows $3.5\times\text{--}3.7\times$ speedup over HotSpot, which comes from its faster linear equation solving. PIRW has $136\times\text{--}209\times$ speedup and $8.53\times\text{--}11.1\times$ storage space reduction over 3D-ICE in cases (*Mix4* and *Stack-1*) with three types of BCs, and certainly has speedup loss in cases (*R4* and *Stack-2*), $1\times\text{--}1.53\times$ speedup, for lacking the absorbing Dirichlet boundary, which can terminate paths in advance. Experimental results verify the accuracy and efficiency of PIRW for mixed BCs.

For pPIRW sharing the same random paths as PIRW, the accuracy, offline CPU time, and offline memory of pPIRW (in parentheses) are similar to those of PIRW. However, the former needs slightly more runtime and memory for storing and calculating trajectory information. The online part of pPIRW, however, needs only $3.46\times\text{--}6.92\mu\text{s}$ in all cases, $2.5\text{e}4\times\text{--}6.7\text{e}6\times$ speedup, and $36.6\times\text{--}42.5\times$ storage space reduction over PIRW. The ultrafast speed and minimal memory consumption make the pPIRW predictor highly suitable for high-speed application scenarios, such as online temperature prediction.

C. Parallelism Analysis

In this experiment, we utilize the same workload as before with the number of parallel CPU cores ranging from 1 to 80, as illustrated in Fig. 12. A customized thread-safe program is leveraged to facilitate multithreaded acceleration. It is observable that as the number of CPU cores increases, the speedup of PIRW maintains a nearly ideal linear growth. The exceptional parallel efficiency of PIRW is attributed to the high level of independence among paths and the similarity in particle behavior within each individual path, which demonstrates notable benefits in large-scale parallel computing.

TABLE II
COMPARISON AMONG TRADITIONAL RANDOM METHOD HRW AND PIRW

Case	ANSYS_H [3]	HRW [22]			PIRW		
	Tem. (°C)	Tem.(°C)	Abs. Err.(°C)	Time(s)	Tem.(°C)	Abs. Err.(°C)	Time(s)
D4	72.197	71.987	0.210	0.11	72.122	0.075	0.10
DP	71.129	71.141	0.012	0.11	71.127	0.002	0.12
R4	93.162	100.133	6.971	3.13	93.010	0.152	23.2
RP	61.574	66.161	4.587	3.10	61.165	0.409	23.3

TABLE III
ACCURACY AND EFFICIENCY OF ANSYS, HotSpot, 3D-ICE, PIRW, AND pPIRW

Case	Methods	Element Size (m)	# Elements	Temperature at hot-spots (°C)	Absolute Error (°C)	Solving Time (s)	Total Time (s)	Speedup	Memory (MB)	Storage Reduction
Mix4	ANSYS_H	6e-5	9.4e6	73.810	-	580	722	-	25143	-
	ANSYS_M	2e-4	6.0e5	73.522	0.288	38	60.0	0.39	1412	0.39
	HotSpot	2e-4	1.1e5	73.250	0.560	80.1	80.5	0.29	1331	0.41
	3D-ICE	2e-4	1.1e5	74.097	0.287	22.9	23.1	1.00	551.8	1.00
	PIRW	-	-	73.735	0.075	-	0.17	136	64.71	8.53
	pPIRW	-	-	73.743	0.067	-	3.46e-6 (0.16[*])	6.68e6	1.53 (66.15[*])	361
Stack-1	ANSYS_H	6e-5	1.64e7	73.740	-	1855	2204	-	62765	-
	ANSYS_M	2.1e-4	1.2e6	73.716	0.024	157	200	0.18	11447	0.06
	HotSpot	2e-4	1.3e5	73.620	0.120	131.3	131.9	0.27	1843.2	0.39
	3D-ICE	2e-4	1.3e5	73.642	0.098	35.3	35.5	1.00	723.5	1.00
	PIRW	-	-	73.750	0.010	-	0.17	209	65.16	11.1
	pPIRW	-	-	73.737	0.003	-	6.92e-6 (0.16[*])	5.13e6	1.78 (68.13[*])	406
R4	ANSYS_H	6e-5	9.6e6	93.162	-	591	770	-	26282	-
	ANSYS_M	2e-4	6.3e5	92.953	0.209	38.3	62.0	0.38	1480	0.37
	HotSpot	2e-4	1.1e5	91.500	1.662	79.9	80.4	0.29	1331	0.41
	3D-ICE	2e-4	1.1e5	93.425	0.263	23.0	23.3	1.00	543.6	1.00
	PIRW	-	-	93.010	0.152	-	23.2	1.00	65.09	8.35
	pPIRW	-	-	93.134	0.028	-	3.46e-6 (26.12[*])	6.73e6	1.53 (66.15[*])	355
Stack-2	ANSYS_H	6e-5	1.7e7	84.604	-	1827	2124	-	62868	-
	ANSYS_M	2.5e-4	8.6e5	84.473	0.131	108	134	0.27	6863	0.11
	HotSpot	2e-4	1.3e5	82.420	2.184	131.8	132.4	0.27	1843.2	0.40
	3D-ICE	2e-4	1.3e5	84.465	0.139	35.3	35.6	1.00	731.1	1.00
	PIRW	-	-	84.729	0.125	-	23.3	1.53	65.21	11.2
	pPIRW	-	-	84.723	0.119	-	6.92e-6 (25.89[*])	5.14e6	1.78 (68.09[*])	411

* The values in parentheses represent the time and memory calculated offline by pPIRW.

TABLE IV
ERROR BETWEEN pPIRW AND CTM

Point	$\ \mathbf{a}\ _2$	$\ \mathbf{a}' - \mathbf{a}\ _2$	$\frac{\ \mathbf{a}' - \mathbf{a}\ _2}{\ \mathbf{a}\ _2}$	Rel. Sum Error of $\mathbf{a}'$	Rel. Sum Error of $\mathbf{r}'$
P_1	1140.5	26.57	2.33%	0.17%	0.006%
P_2	1140.4	26.65	2.34%	0.18%	0.006%
P_3	1140.4	26.65	2.33%	0.18%	0.006%

D. Relationship Between CTM and pPIRW

We utilize MatEx [5], a CTM approach, to construct and export the thermal resistor matrix, and the perform matrix inversion with eigen library [34]. A piece of 1.6 mm $\times$ 1.6 mm is vertically cut from the 2 cm $\times$ 2 cm case in Fig. 11 for verifying the consistency between pPIRW and CTM with high accuracy. The heatless regions above and below are 1.6 mm $\times$ 1.6 mm $\times$ 0.5 mm, evenly discretized into 16 $\times$ 16 $\times$ 25 grids, and the heat-source region is 1.6 mm $\times$ 1.6 mm $\times$ 0.1 mm, evenly discretized into 16 $\times$ 16 $\times$ 5 = 1280 grids. Vector $\mathbf{a}$ with a dimension of 1280 will be a part of a row in the inverse matrix $\mathbf{G}^{-1}$ from CTM as shown in Fig. 9, and vector $\mathbf{a}'$ with the same dimension is calculated by pPIRW with the sufficient number of paths $N = 1e6$. Three points P_1 – P_3 in the middle layer of the heat-source region as shown in Fig. 13(a) are selected.

Comparison results of vectors $\mathbf{a}$ from the CTM and the vectors $\mathbf{a}'$ from the pPIRW in P_1 – P_3 are shown in Table IV, where $\|\cdot\|_2$ is the L2 norm. The relative sum error of $\mathbf{a}'$ is defined as $(\|\sum \mathbf{a} - \sum \mathbf{a}'\|_2 / \|\sum \mathbf{a}\|_2)$, which characterizes the final temperature error at the point according to the definition of vector multiplication. A similar definition applies to $\mathbf{r}'$. The specific differences in the elements of $\mathbf{a}$ and $\mathbf{a}'$ are illustrated in Fig. 13(b). The 2.34% relative L2 norm error and 0.18%

and 0.0006% relative sum error verify the consistency between the pPIRW and the CTM.

E. Application of pPIRW Predictor in Proactive Thermal Management

This section evaluates the effectiveness of applying pPIRW predictor in the proactive thermal management architecture presented.

1) *Experimental Setting of Thermal Management*: For modeling more realistic packaging chips, the bottom heatless cuboid region's thickness, size, and thermal conductivity are set to 100 μ m, 1.6 cm $\times$ 1.6 cm, and 50 W/(K·m) in Fig. 11, respectively. The ambient temperature is 45 °C, the top surface is the Robin boundary, and the bottom surface is the adiabatic Neumann boundary. Case *TM-1* and *TM-2* have different h on the top Robin boundary, i.e., 12 500 or 10 000 W/(K·m²) as shown in Table I, for different cooling conditions. The core-level power-map, *16core* in Fig. 11 assuming uniform power distribution on each core, is derived from allocated synthetic workload in time series.

We compare the pPIRW predictor with the empirical RM predictor and the CTM predictor under the same thermal management strategy illustrated in Section IV-D. The 16-core processor is evenly organized into 2 $\times$ 2 clusters, and each task is scheduled to run with four-core parallelism on a cluster. Scheduling tasks include *blackscholes*, *bodytrack*, *streamcluster*, *cannal*, *x264*, *swaptions*, and *dedup* from PARSEC v2.1 [35] benchmark suite. Four V/f levels (2.0 GHz, 1.2 V), (1.8 GHz, 1.1 V), (1.5 GHz, 1.0 V), and (1.0 GHz, 0.9 V) on each cluster are adopted.

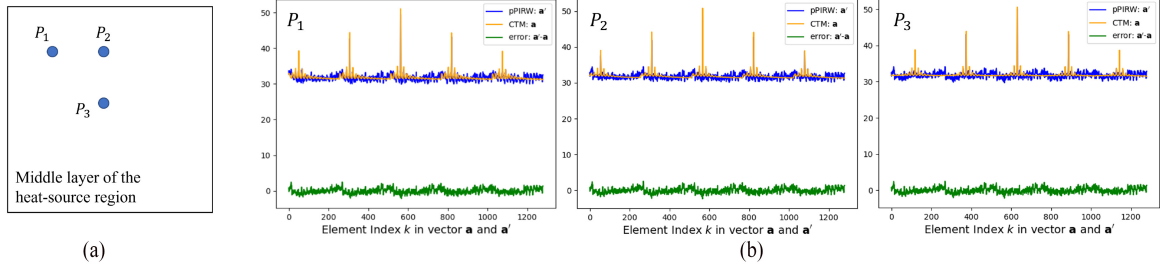


Fig. 13. Verification of the consistency between pPIRW and CTM. (a) Top view of points' locations. (b) Specific differences in the 1280 elements.

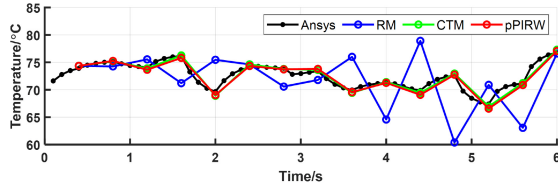


Fig. 14. Accuracy comparison among RM, CTM, and pPIRW predictors with ANSYS as the golden solution.

The total simulation time of transient thermal analysis is 10 s. The *Golden* results are from medium-accuracy ANSYS with fine enough element size $2e-4m$ and time step $\Delta t_1 = 0.1$ s to capture transient effects. The time step of the power-map change as well as the trigger of proactive thermal management is $\Delta t = 0.4$ s.

To train the RM, we simulate a full-time transient analysis by ANSYS with randomly selecting V/f levels, and obtain 400 points (25 time steps $\times$ 16 cores) with true temperatures. From (7), temperature $\tilde{T}_i$ of the RM prediction is dependent on T_{i-1} and T_{i-2} , the previous true temperatures read from sensors [31]. In our simulation, the golden temperatures from ANSYS are fed into the RM predictor.

Without loss of generality, we randomly choose some locations on each core and regard them as points of interest, as shown in the *16core* power-map in Fig. 11. Note that, the CTM needs the fine heat-source discretization of $40 \times 40 \times 25$ to capture temperatures at arbitrary locations.

2) *Accuracy Comparison of Different Predictors:* Fig. 14 compares the accuracy of RM, CTM, and pPIRW predictors on a core during a thermal management session. From Fig. 14, one can clearly see that pPIRW and CTM are very close to the golden ANSYS, while RM has significant deviations. The same conclusions can be drawn from Table V, where the average and maximal absolute errors of pPIRW and CTM are less than 0.27°C and 1.15°C respectively, but those of RM are more than 3.82°C and 19.6°C .

Fundamentally, the RM is a fitting method designed for the case with only one temperature sensor on the cluster [31]. In the cases of multiple sensors, this simple model tends to underfit. Therefore, even if an error correction mechanism [31] is introduced, the accuracy and stability of predictions are still difficult to guarantee. On the contrary, both CTM and pPIRW are basically accurate physical models with no requirement of sensors, and thus stable and insensitive to sensor accuracy.

3) *Effectiveness Comparison of Thermal Management With Different Predictors:* Table V shows the performance comparison of thermal management with different predictors. In

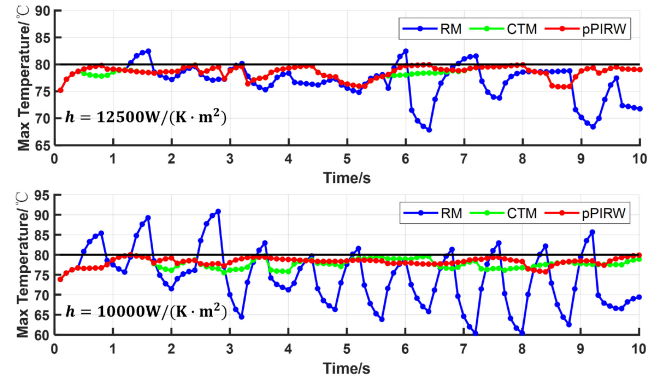


Fig. 15. Maximum actual temperatures over time with different prediction methods.

these cases, the goal of thermal management is to maximize frequency without exceeding the temperature threshold. As indicated in Table V, pPIRW maintains the highest average frequency (1.844 GHz in *TM-1* and 1.675 GHz in *TM-2*) with no conflicts under all heat dissipation scenarios, and CTM also achieves similar good thermal management results. RM's average frequencies, however, are 1.688 and 1.317 GHz with 20 and 74 thermal conflicts in cases *TM-1* and *TM-2*, respectively. A similar but more intuitive conclusion can be observed from Fig. 15. Both pPIRW and CTM effectively keep the maximum temperatures as close to the threshold (80°C) as possible, whereas RM's maximum temperatures are either significantly below the threshold or result in a substantial amount of conflicts. Therefore, for an effective thermal management system, a high-precision thermal predictor seems indispensable.

4) *Online Speed Comparison With Different Predictors:* The online speed comparisons of invoking a thermal management and a temperature prediction are listed in the columns of "One-step Management Time" and "Prediction Time" of Table V, respectively.

Although RM only requires $0.12 \mu\text{s}$ for a temperature prediction, its thermal management needs to access temperature sensors with a time of $586.5 \mu\text{s}$ [31] at every time step, which restricts its speed of one-step thermal management. Both CTM and pPIRW need similar $0.15 \mu\text{s}$ for a temperature prediction and only $3.05\text{--}4.42 \mu\text{s}$ for a one-step management, $134\times$ speedup over RM, for no need to access temperature sensors.

5) *Offline Speed and Memory Comparison Between pPIRW and CTM:* With fine discretization, the CTM is able to achieve high accuracy for hot-spots located at any site, but this comes

TABLE V
THERMAL MANAGEMENT PERFORMANCE WITH DIFFERENT PREDICTORS

Case	Predictors	# Thermal Conflicts	Avg. Freq. (GHz)	Avg. Abs. Error (°C)	Max Abs. Error (°C)	One-step Management Time (μs)	Prediction Time (μs)	Offline Time (s)	Offline Memory (GB)
TM-1	RM	20	1.688	3.826	19.64	590.1*	0.12	0.17	0.97
	CTM	0	1.838	0.168	0.932	3.09	0.15	14673	55.7
	pPIRW	0	1.844	0.255	1.149	3.05	0.15	502	0.09
TM-2	RM	74	1.317	6.269	26.42	591.7*	0.12	0.16	0.97
	CTM	1	1.666	0.216	1.042	4.40	0.15	14690	55.7
	pPIRW	0	1.675	0.268	0.962	4.42	0.15	498	0.09

* Include the time to read thermal sensors from [34].

at the cost of CPU time and memory resources. With the heat-source region grid of $40 \times 40 \times 25$, CTM needs 14 673 s CPU time and 55.7 GB memory, while pPIRW needs only 502 s and 90 MB for the offline calculation, $29.2 \times$ speedup and $634 \times$ storage reduction respectively, as shown in the final two columns of Table V.

The improvement of speed and memory stems from the fact that the stochastic pPIRW avoids global discretization, linear equation system formulation, and time consuming inversion of large matrices required by deterministic methods. In the offline phase, the time complexity of CTM is $O(n^3)$, where n is degrees of freedom, while that of pPIRW is nearly constant at $O(cm)$, where m is the time for a site by stochastic methods, and c is the number of hot-spot sites. The speed gap may be substantial enough for large-scale enterprise cases.

Particularly, the inverse of a sparse matrix is typically not sparse. Directly inverting and storing a massive sparse matrix $\mathbf{G}$ by CTM is usually infeasible. Even worse, only limited elements of $\mathbf{G}^{-1}$ are actually needed, making the full inversion of $\mathbf{G}$ cumbersome and inefficient.

These advantages of pPIRW in speed and memory, discretization-free, and inherent parallelism will be even more pronounced in problems involving large-scale chips and complex geometries.

VI. CONCLUSION

In this article, we propose a path integral random walk stochastic solver for 3-D steady-state thermal analysis with mixed BCs. Owing to the strictly correct calculation of local time and the introduction of the Feynman–Kac functional $\hat{e}_c(t_j)$, our PIRW solver is capable of accurately handling thermal analysis problems with mixed BCs with negligible error within 0.5°C . Meanwhile, the high parallelism of stochastic methods enables the PIRW solver to run efficiently, achieving $136 \times$ – $209 \times$ speedup and $8.53 \times$ – $11.1 \times$ storage space reduction with all three kinds of BCs at a single point.

Based on the PIRW, we further propose an accurate stochastic temperature predictor pPIRW for proactive thermal management. By precalculating the attenuated factor of Robin BCs offline, the temperatures of hot-spots can be obtained online through simple vector multiplication. The pPIRW achieves more than $2.5e4 \times$ – $6.7e6 \times$ speedup and $36.6 \times$ – $42.5 \times$ storage space reduction over PIRW without loss of accuracy. We reveal and validate that pPIRW essentially calculates the partial inverse of the matrix $\mathbf{G}$ constructed in the CTM. After integrating pPIRW into a proactive thermal management framework, pPIRW can control temperatures below the threshold while maximizing performance. Compared to

CTM, pPIRW achieves $29.2 \times$ acceleration and $634 \times$ memory reduction during the offline precalculation stage.

REFERENCES

- [1] S.-C. Lin and K. Banerjee, “Thermal challenges of 3D ICs,” in *Wafer Level 3-D ICs Process Technology*. Boston, MA, USA: Springer, 2008, pp. 1–26. [Online]. Available: https://link.springer.com/chapter/10.1007/978-0-387-76534-1_14
- [2] “Ansys.” 2018. [Online]. Available: <https://www.ansys.com/>
- [3] “COMSOL.” 2022. [Online]. Available: <https://www.comsol.com/>
- [4] H. Sultan, A. Chauhan, and S. R. Sarangi, “A survey of chip-level thermal simulators,” *ACM Comput. Surveys*, vol. 52, no. 2, pp. 1–35, 2019.
- [5] S. Pagani, J.-J. Chen, M. Shafique, and J. Henkel, “MatEx: Efficient transient and peak temperature computation for compact thermal models,” in *Proc. Design, Autom. Test Europe Conf. Exhibit. (DATE)*, 2015, pp. 1515–1520.
- [6] Z. Yuan, P. Shukla, S. Chetoui, S. Nemtsov, S. Reda, and A. K. Coskun, “PACT: An extensible parallel thermal simulator for emerging integration and cooling technologies,” *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 41, no. 4, pp. 1048–1061, Apr. 2022.
- [7] K. Skadron, M. R. Stan, K. Sankaranarayanan, W. Huang, S. Velusamy, and D. Tarjan, “Temperature-aware microarchitecture: Modeling and implementation,” *ACM Trans. Archit. Code Optim.*, vol. 1, no. 1, pp. 94–125, 2004.
- [8] A. Sridhar, A. Vincenzi, M. Ruggiero, T. Brunschweiler, and D. Atienza, “3D-ICE: Fast compact transient thermal modeling for 3D ICs with inter-tier liquid cooling,” in *Proc. IEEE/ACM Int. Conf. Comput.-Aided Design (ICCAD)*, 2010, pp. 463–470.
- [9] A. Sridhar, A. Vincenzi, M. Ruggiero, T. Brunschweiler, and D. Atienza, “Compact transient thermal model for 3D ICs with liquid cooling via enhanced heat transfer cavity geometries,” in *Proc. 16th Int. Workshop Thermal Investigations ICs Syst. (THERMINIC)*, 2010, pp. 1–6.
- [10] F. Terraneo, A. Leva, W. Fornaciari, M. Zapater, and D. Atienza, “3D-ICE 3.0: Efficient nonlinear MPSoC thermal simulation with pluggable heat sink models,” *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 41, no. 4, pp. 1062–1075, Apr. 2022.
- [11] R. Ranade, H. He, J. Pathak, N. Chang, A. Kumar, and J. Wen, “A thermal machine learning solver for chip simulation,” in *Proc. ACM/IEEE Workshop Mach. Learn. CAD*, 2022, pp. 111–117.
- [12] O. Hennigh et al., “NVIDIA SimNet™: An AI-accelerated multi-physics simulation framework,” in *Proc. Int. Conf. Comput. Sci.*, 2021, pp. 447–461.
- [13] Z. Liu et al., “DeepOHeat: Operator learning-based ultra-fast thermal simulation in 3D-IC design,” in *Proc. 60th ACM/IEEE Design Autom. Conf. (DAC)*, 2023, pp. 1–6.
- [14] J.-P. Morillon, “Numerical solutions of linear mixed boundary value problems using stochastic representations,” *Int. J. Numer. Methods Eng.*, vol. 40, no. 3, pp. 387–405, 1997.
- [15] G. E. Forsythe and R. A. Leibler, “Matrix inversion by a Monte Carlo method,” *Math. Comput.*, vol. 4, no. 31, pp. 127–129, 1950.
- [16] R. Iverson, Y. Le Coz, B. Kleveland, and S. Wong, “A multi-scale random-walk thermal-analysis methodology for complex IC-interconnect systems,” in *Proc. Int. Conf. Simul. Semicond. Processes Devices*, 2000, pp. 84–86.
- [17] J. Guo, S. Dong, and S. Goto, “Random walk algorithm for large thermal RC network analysis,” in *Proc. IEEE 8th Int. Conf. ASIC*, 2009, pp. 771–774.
- [18] E. Wong and S. K. Lim, “3D floorplanning with thermal vias,” in *Proc. Design Autom. Test Europe Conf.*, vol. 1, 2006, pp. 1–6.
- [19] Y. Huang, Q. Zhou, Y. Cai, and H. Yan, “A thermal-driven force-directed floorplanning algorithm for 3D ICs,” in *Proc. 11th IEEE Int. Conf. Comput.-Aided Design Comput. Graph.*, 2009, pp. 497–502.

- [20] Y. Liang, W. Yu, and H. Qian, "A hybrid random walk algorithm for 3-D thermal analysis of integrated circuits," in *Proc. 19th Asia South Pacific Design Autom. Conf. (ASP-DAC)*, 2014, pp. 849–854.
- [21] P. Maffezzoni and A. Brambilla, "Analysis of substrate coupling by means of a stochastic method," *IEEE Electron Device Lett.*, vol. 23, no. 6, pp. 351–353, Jun. 2002.
- [22] L. Yang, C. Ding, C. Yan, D. Zhou, and X. Zeng, "A high-precision stochastic solver for steady-state thermal analysis with fourier heat transfer robin boundary conditions," in *Proc. 41st ACM/IEEE Int. Conf. Comput.-Aided Design (ICCAD)*, 2022, pp. 1–9.
- [23] P. Hsu, *Reflecting Brownian Motion, Boundary Local Time and the Neumann Problem*. Stanford, CA, USA: Stanford Univ., 1984.
- [24] V. G. Papanicolaou, "The probabilistic solution of the third boundary value problem for second order elliptic equations," *Probab. Theory Rel. Fields*, vol. 87, no. 1, pp. 27–77, 1990.
- [25] Y. Zhou, W. Cai, and E. Hsu, "Computation of the local time of reflecting Brownian motion and the probabilistic representation of the Neumann problem," *Commun. Math. Sci.*, vol. 15, no. 1, pp. 237–259, 2017.
- [26] K. K. Sabelfeld, *Monte Carlo Methods in Boundary Value Problems*, vol. 274. Heidelberg, Germany: Springer, 1991.
- [27] A. K. Singh, S. Dey, K. McDonald-Maier, K. R. Basireddy, G. V. Merrett, and B. M. Al-Hashimi, "Dynamic energy and thermal management of multi-core mobile platforms: A survey," *IEEE Design Test*, vol. 37, no. 5, pp. 25–33, Oct. 2020.
- [28] S. Pagani, P. S. Manoj, A. Jantsch, and J. Henkel, "Machine learning for power, energy, and thermal management on multicore processors: A survey," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 39, no. 1, pp. 101–116, Jan. 2020.
- [29] A. K. Coskun, T. S. Rosing, and K. C. Gross, "Utilizing predictors for efficient thermal management in multiprocessor SoCs," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 28, no. 10, pp. 1503–1516, Oct. 2009.
- [30] Y. Ge, Q. Qiu, and Q. Wu, "A multi-agent framework for thermal aware task migration in many-core systems," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 20, no. 10, pp. 1758–1771, Oct. 2012.
- [31] E. W. Wächter, C. De Bellefroid, K. R. Basireddy, A. K. Singh, B. M. Al-Hashimi, and G. Merrett, "Predictive thermal management for energy-efficient execution of concurrent applications on heterogeneous multicores," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 27, no. 6, pp. 1404–1415, Jun. 2019.
- [32] A. V. Skorokhod, "Stochastic equations for diffusion processes in a bounded region," *Theory Probab. Appl.*, vol. 6, no. 3, pp. 264–274, 1961.
- [33] C. Ding, Y. Zhou, W. Cai, X. Zeng, and C. Yan, "A path integral Monte Carlo (PIMC) method based on Feynman-Kac formula for electrical impedance tomography," *J. Comput. Phys.*, vol. 476, Mar. 2023, Art. no. 111862.
- [34] "Eigen C++ template library," Eigen, 2021. [Online]. Available: <http://eigen.tuxfamily.org/>
- [35] C. Bienia, S. Kumar, J. P. Singh, and K. Li, "The PARSEC benchmark suite: Characterization and architectural implications," in *Proc. 17th Int. Conf. Parallel Archit. Compilation Techn.*, 2008, pp. 72–81.



Zhixuan Dong received the B.S. degree in microelectronics from Fudan University, Shanghai, China, in 2022, where she is currently pursuing the Ph.D. degree with the State Key Laboratory of Integrated Chips and Systems, Microelectronics Department.

Her current research interests include thermal analysis and multiphysics modeling and simulation.



Longlong Yang received the B.S. and M.S. degrees in microelectronics from Fudan University, Shanghai, China, in 2020 and 2023, respectively.

He is currently working with Shanghai UniVista Industrial Software Group Company Ltd., Shanghai.



Cuiyang Ding received the B.S. and Ph.D. degrees in microelectronics from Fudan University, Shanghai, China, in 2018 and 2023, respectively.

He is currently working with Huada Emphyrean Software Company Ltd., Shanghai.



Changhao Yan (Member, IEEE) received the B.E. and M.E. degrees from Huazhong University of Science and Technology, Wuhan, China, in 1996 and 2002, respectively, and the Ph.D. degree in computer science from Tsinghua University, Beijing, China, in 2006.

He is currently a Full Professor with the School of Microelectronics, Fudan University. His current research interests include parasitic parameter extraction of interconnects, parallel algorithms for large scale computation, design for manufacturability, the

robust analysis of circuits, and AI and machine learning in medicine.



Zhaori Bi (Member, IEEE) received the B.Eng. degree in electronic information engineering from Wuhan University of Technology, Wuhan, China, in 2011, and the M.S. and Ph.D. degrees in electrical engineering and computer engineering from the University of Texas at Dallas, Richardson, TX, USA, in 2013 and 2017, respectively.

He is currently an Assistant Professor with Fudan University, Shanghai, China. His current research interests include mixed-signal system-on-a-chip design, circuit performance optimization, and

applications in medical AI.



Sheng-Guo Wang (Life Senior Member, IEEE) received the B.S. degree in electrical engineering from the University of Science and Technology of China, Beijing, China, in 1967, the M.S. degree in electrical engineering from the University of Science and Technology of China, Hefei, China, in 1981, and the Ph.D. degree in electrical and computer engineering from the University of Houston, Houston, TX, USA, in 1994.

He is currently a Professor with the University of North Carolina at Charlotte, Charlotte, NC, USA.

His current research interests include systems, circuits, control, modeling, robust control, structural control, intelligent control, GPS, communications, computer networks, mathematical, and numerical analysis, algorithms, and applications.



Dian Zhou (Senior Member, IEEE) received the B.S. degree in physics and the M.S. degree in electrical engineering from Fudan University, Shanghai, China, in 1982 and 1985, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Illinois at Urbana-Champaign, Champaign, IL, USA, in 1990.

He joined the University of North Carolina at Charlotte, Charlotte, NC, USA, as an Assistant Professor in 1990, where he became an Associate Professor in 1995. He joined the University of Texas

at Dallas, Richardson, TX, USA, as a Full Professor in 1999. His research interests include high-speed VLSI systems, CAD tools, mixed-signal ICs, and algorithms.



Xuan Zeng (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Fudan University, Shanghai, China, in 1991 and 1997, respectively.

She is currently a Full Professor with the School of Microelectronics, Fudan University. Her current research interests include analog circuit modeling and synthesis, design for manufacturability, high-speed interconnect analysis and optimization, and circuit simulation.

Prof. Zeng received the Changjiang Distinguished Professor with the Ministry of Education Department of China in 2014, and the Chinese National Science Funds for Distinguished Young Scientists in 2011.